



The Damaging Effects of Smoking – and How to Quit

【【生词双高亮】 TED科普：烟如何侵蚀你的身体？ 】

Summary: Smoking introduces over 5,000 chemicals that damage teeth, impair smell, and harm lungs by destroying cilia. Carbon monoxide reduces oxygen in blood, while nicotine triggers addictive dopamine release. Long-term effects include cancer risks, vascular damage, and reproductive issues. Quitting brings immediate benefits: heart rate normalizes in 20 minutes, lung function improves within months, and cancer risks drop significantly over years. Support tools like nicotine therapy and counseling aid cessation.

摘要：吸烟时5000多种化学物质会侵蚀牙齿、损伤嗅觉并破坏肺部纤毛。一氧化碳降低血氧含量，尼古丁刺激多巴胺释放导致成瘾。长期吸烟可能引发癌症、血管病变和生育问题。戒烟后20分钟心率即恢复正常，数月后肺部功能改善，数年内癌症风险显著下降。尼古丁替代疗法和心理辅导等工具可有效辅助戒烟。





HOW DO CIGARETTES



双语字幕 | 生词高亮 | 智能文稿

🕒 Estimated Reading Time: 7 min.

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🎵 Cigarettes aren't good for us.

香烟对我们没有好处。

🎵 That's hardly news--we've known about the dangers of smoking for decades.

这算不上新闻——几十年来我们一直清楚吸烟的危害。

🎵 But how exactly do cigarettes harm us?

但香烟究竟如何伤害我们？

▶ Let's look at what happens as their ingredients make their way through our bodies, and how we benefit physically when we finally give up smoking.

让我们看看当香烟成分在体内扩散时会发生什么，以及最终戒烟时身体如何受益。

▶ With each inhalation, smoke brings its more than 5,000 chemical substances into contact with the body's tissues.

每次吸入时，烟雾中超过5000种化学物质都会与人体组织接触。

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▶ From the start, tar, a black, resinous material, begins to coat the teeth and gums, damaging tooth enamel, and eventually causing decay.

一开始，#焦油#——一种黑色#树脂状#物质——就会覆盖牙齿和牙龈，破坏牙釉质并最终导致蛀牙。

▶ Over time, smoke also damages nerve-endings in the nose, causing loss of smell.

随着时间的推移，烟雾还会损害鼻子里的神经末梢，导致嗅觉丧失。

▶ Inside the airways and lungs, smoke increases the likelihood of infections, as well as chronic diseases like bronchitis and emphysema.

在气道和肺部，烟雾会增加感染#可能性#以及支气管炎、肺气肿等#慢性#疾病风险。

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▶ It does this by damaging the cilia, tiny hairlike structures whose job it is to keep the airways clean.

其原理是破坏纤毛——这些负责保持气道清洁的微小毛发状结构。

▶ It then fills the alveoli, tiny air sacs that enable the exchange of oxygen and carbon dioxide between the lungs and blood.

随后烟雾会填满肺泡——这些微小气囊负责实现肺部与血液间的#氧气#和二氧化碳#交换#。

▶ A toxic gas called carbon monoxide crosses that membrane into the blood, binding to hemoglobin and displacing the oxygen it would usually have transported around the body.

有毒的一氧化碳气体会穿过肺泡膜进入血液，与血红蛋白结合，取代通常由血红蛋白输送的氧气。

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▶ That's one of the reasons smoking can lead to oxygen deprivation and shortness of breath.

这就是吸烟可能导致#缺氧#和呼吸急促的原因之一。

▶ Within about 10 seconds, the bloodstream carries a stimulant called nicotine to the brain, triggering the release of dopamine and other neurotransmitters including endorphins that create the pleasurable sensations which make smoking highly addictive.

约10秒内，血液会将名为尼古丁的兴奋剂输送到大脑，触发多巴胺和内啡肽等神经递质的释放，产生愉悦感，这正是吸烟高度成瘾的原因。

▶ Nicotine and other chemicals from the cigarette simultaneously cause constriction of blood vessels and damage their delicate endothelial lining, restricting blood flow.

#尼古丁#和香烟中的其他化学物质会#同时#导致血管收缩并破坏脆弱的内皮#内膜#，从而限制血流。

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▶ These vascular effects lead to thickening of blood vessel walls and enhance blood platelet stickiness, increasing the likelihood that clots will form and trigger heart attacks and strokes.

这些血管效应会导致血管壁增厚、血小板黏性增强，增加形成血栓并引发心脏病和中风的风险。

▶ Many of the chemicals inside cigarettes can trigger dangerous mutations in the body's DNA that make cancers form.

香烟中的许多化学物质会#引发#人体DNA的危险突变，从而导致癌症形成。

▶ Additionally, ingredients like arsenic and nickel may disrupt the process of DNA repair, thus compromising the body's ability to fight many cancers.

此外，砷和#镍#等成分可能#破坏#DNA修复过程，从而削弱身体对抗多种癌症的能力。

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▶ In fact, about one of every three cancer deaths in the United States is caused by smoking.

事实上，美国每三例癌症死亡中约有一例是由吸烟引起。

▶ And it's not just lung cancer.

而且不仅仅是肺癌。

▶ Smoking can cause cancer in multiple tissues and organs, as well as damaged eyesight and weakened bones.

吸烟会导致多种组织和器官的癌症，还会损害#视力#并削弱骨骼。

▶ It makes it harder for women to get pregnant.

它会使女性更难怀孕。

▶ And in men, it can cause erectile dysfunction.

对男性而言，它可能导致勃起#功能障碍#。

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▶ But for those who quit smoking, there's a huge positive upside with almost immediate and long-lasting physical benefits.

但对戒烟者而言，身体几乎能立即获得持久#的积极效益。

▶ Just 20 minutes after a smoker's final cigarette, their heart rate and blood pressure begin to return to normal.

吸烟者最后一支烟仅20分钟后，心率和血压就开始恢复正常。

▶ After 12 hours, carbon monoxide levels stabilize, increasing the blood's oxygen-carrying capacity.

12小时后，一氧化碳水平#趋于稳定#，血液携#氧#能力提升。

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▶ A day after ceasing, heart attack risk begins to decrease as blood pressure and heart rates normalize.

停止吸烟一天后，随着血压和心率正常化，心脏病发作风险开始下降。

🎧 After two days, the nerve endings responsible for smell and taste start to recover.

两天后，负责嗅觉和味觉的神经末梢开始恢复。

🎧 Lungs become healthier after about one month, with less coughing and shortness of breath.

约一个月后肺部变得更健康，咳嗽和呼吸急促症状减轻。

🎧 The delicate hair-like cilia in the airways and lungs start recovering within weeks, and are restored after 9 months, improving resistance to infection.

气道和肺部纤细的毛发状纤毛在数周内开始恢复，9个月后完全修复，从而增强抗感染能力。

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🎧 By the one-year anniversary of quitting, heart disease risk plummets to half as blood vessel function improves.

戒烟满一周年时，随着血#管#功能改善，心脏病风险骤降至一半。

🎧 Five years in, the chance of a clot forming dramatically declines, and the risk of stroke continues to reduce.

五年后，形成#血栓#的几率显著下降，中风风险持续降低。

🎧 After ten years, the chances of developing fatal lung cancer go down by 50%, probably because the body's ability to repair DNA is once again restored.

十年后，致命肺癌的发病几率下降50%，这可能是因为人体修复DNA的能力再次恢复。

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🎧 Fifteen years in, the likelihood of developing coronary heart disease is essentially the same as that of a non-smoker.

十五年后，患冠心病的#可能性#基本与不吸烟者持平。

🎧 There's no point pretending this is all easy to achieve.

没有必要假装这一切都很容易实现。

🎧 Quitting can lead to anxiety and depression, resulting from nicotine withdrawal.

戒烟可能导致焦虑和抑郁，这是#尼古丁#戒断引起的。

🎧 But fortunately, such effects are usually temporary.

但#幸运的是#，这些影响通常是暂时的。

🎧 And quitting is getting easier, thanks to a growing arsenal of tools.

由于戒烟#手段#日益丰富，戒烟正变得越来越容易。

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🎧 Nicotine replacement therapy through gum, skin patches, lozenges, and sprays may help wean smokers off cigarettes.

通过#口香糖#、皮肤贴片、含片和喷雾进行的#尼古丁##替代#疗法可以帮助#戒除#吸烟习惯。

🎧 They work by stimulating nicotine receptors in the brain and thus preventing withdrawal symptoms, without the addition of other harmful chemicals.

其原理是刺激大脑中的#尼古丁#受体，从而避免戒断症状，且不添加其他有害化学物质。

🎧 Counselling and support groups, cognitive behavioral therapy, and moderate intensity exercise also help smokers stay cigarette-free.

咨询辅导、互助小组、#认知#行为疗法和#中等##强度#锻炼也有助于吸烟者保持无烟状态。

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🎧 That's good news, since quitting puts you and your body on the path back to health.

这是个好消息，因为戒烟能让你和你的身体重回健康之路。

